



Instructions For Use

Valid as of 18.08.2022

AKLIDES® CytoBead® RPGN

REF 4281

Indirect immunofluorescence assay
for the determination of IgG antibodies
in human serum against neutrophil cytoplasmic
antigens (ANCA), the glomerular basement
membrane (GBM) and double stranded
deoxyribonucleic acid (dsDNA)



Medipan GmbH
Ludwig-Erhard-Ring 3
15827 Blankenfelde-Mahlow OT Dahlewitz
Germany

Phone: +49 33708 4417 0
Fax: +49 33708 4417 25

info@medipan.de
www.medipan.de



1 Intended Purpose

The AKLIDES® CytoBead® ANCA is an immunofluorescence assay (IFA) for the semi-quantitative determination of IgG antibodies in human serum against neutrophil cytoplasmic antigens (ANCA), and quantitative determination of IgG antibodies against proteinase 3 (PR3), myeloperoxidase (MPO), glomerular basal membrane (GBM) and double stranded deoxyribonucleic acid (dsDNA).

The AKLIDES® CytoBead® RPGN is intended as an aid in the in the diagnosis of systemic vasculitis as well as autoimmune renal disorders in conjunction with other clinical and laboratory findings.

The immunoassay is designed for manual processing and automated imaging using the AKLIDES® or akiron® system.

The immunoassay is designed for professional *in vitro* diagnostic use only.

2 Diagnostic Relevance

Anti-neutrophil cytoplasmic antibodies (ANCA) are a group of autoantibodies directed against cytoplasmic components of neutrophilic granulocytes and are associated with inflammatory vascular diseases.

According to their immunofluorescence pattern on ethanol-fixed human granulocytes ANCA have been classified into cytoplasmic (cANCA) and perinuclear (pANCA) antibodies. Antibodies to proteinase 3 (PR3), the main autoantigen of cANCA pattern, are highly specific for Wegener Granulomatosis, a granulomatous disease of nasopharynx, kidney and lung. Antibodies to myeloperoxidase (MPO) are causing a majority of pANCA patterns and have been reported in a variety of vasculitides (Microscopic polyangitis, Churg-Strauss Syndrome, Polyarteritis nodosa) and glomerulonephritides (rapidly progressive glomerulonephritis).

Goodpasture syndrome is a medical emergency with a high fatality rate if not treated. The syndrome is characterized by glomerulonephritis, pulmonary haemorrhage and antibody formation against glomerular basement membrane (GBM). The antibodies are targeted to a GBM component, the collagen IV and specifically to the NC1 domain of the $\alpha 3$ chain (Goodpasture antigen). Anti-GBM antibodies are the primary pathogenic autoantibodies, binding along the glomerular basement membrane and inducing glomerulonephritis in all patients with

Goodpasture syndrome. Their determination allows to differentiate the syndrome from other causes of glomerular nephritis and pulmonary haemorrhage.

Lupus nephritis is an inflammation of the kidneys that occurs as a consequence of systemic lupus erythematosus (SLE). Histopathologically, lupus nephritis shows glomerulonephritis with immune complex deposits as a result of the formation of autoantibodies. Other pathological changes include tubulointerstitial nephritis and vascular changes with immune complex deposits in the vessels and microangiopathy. Antibodies against dsDNA are important for the diagnosis of SLE.

3 Test Principle

The CytoBead® is an innovative indirect immunofluorescence immunoassay for the determination of specific antibodies. Tissue sections or cells containing the antigen of interest as well as additional beads coated with specific antigens are immobilized on slides. If specific antibodies are present in the patient's sample, they bind to the antigen. A secondary antibody conjugated with fluorescein-isothiocyanate (FITC) detects the generated immune complex. The slides are examined using a fluorescence microscope. A specific fluorescent staining pattern demonstrates the presence of specific antibodies in the patient's sample.

4 Test Components

Component	Description
Substrate slide A SLIDE 8 6 pieces	6 slides (ready-to-use), 8 wells per slide, each well coated with cryostat tissue sections of human ethanol fixed granulocytes as well as with beads coated with Proteinase 3 (PR3), Myeloperoxidase (MPO), Glomerular basement membrane (GBM) and dsDNA
Calibrator 0 – 3 CAL 4 x 2 mL	Colored dilutions of human serum (ready-to-use; contains sodium azide) The antibody activities are indicated on the quality control certificate.
Negative control CI CONTROL 1 x 2 mL	Dilution of human serum (ready-to-use; contains sodium azide)
Positive control CII CONTROL 1 x 2 mL bead control	Dilution of human serum (ready-to-use; contains sodium azide)
Positive control CIII CONTROL 1 x 2 mL cell control	Dilution of human serum (ready-to-use; contains sodium azide)
Sample diluent B DIL 1 x 20 mL	Solution (ready-to-use; contains sodium azide)
Wash buffer/ Diluent C WASHB 3 x 10 g	PBS buffer (dry substance for 3 x 1000 mL)
Conjugate IgG D CONJ 1 x 10 mL	Polyclonal anti-human IgG antibody conjugated to FITC (ready-to-use; contains DAPI and sodium azide)
Mounting medium G MOUNT 1 x 3 mL	PBS buffered solution with glycerol and antifading reagent (ready-to-use; contains sodium azide)
Blotting templates F TEMPL 6 pieces	Blotting templates
Coverslips H COVER 12 pieces	Coverslips
QC Certificate 1 piece	-
Instructions for Use 1 piece	-

5 Materials required but not provided

- Common laboratory equipment

- Precision pipettes (5 – 1000 µL) and disposable tips
- Graduated cylinders (100 – 1000 mL)
- Sample tubes for the preparation of dilutions and racks
- Vortex mixer or other rotators
- Moist chamber
- Aspiration pump
- Coplin jars or staining dishes with slide racks
- AKLIDES® or akiron® system or equivalents for imaging
- Adsorbent paper or paper towel
- Distilled or de-ionized water

1 Storage and Stability

Upon receipt, all test components must be stored at 2 °C to 8 °C, preferably in the original kit box. If stored properly in their original containers, all components are stable until their expiry date. After opening, all kit components are stable for at least 2 months, if stored properly.

2 General Information

This product is for *in vitro* diagnostic use only. The instructions for use must be carefully read before use. They are valid only for the present product with the given composition and must be strictly followed to ensure reliable test results. Deviations can lead to erroneous test results. Components must not be exchanged by test reagents of different lots or of other manufacturers.

Contamination of reagents must be avoided by use of aseptic techniques when removing aliquots from the vials. After use, reagent vials must be tightly closed with their corresponding caps.

Cross-contamination of samples or reagents can lead to inconsistent test results and must be avoided by use of consistent pipetting techniques.

Exposure of reagents to strong light must be avoided throughout the entire test procedure and storage.

Insufficient washing will result in poor precision and elevated measurement signals. After each washing step any residual fluid has to be removed completely.

3 Preparation

3.1 Preparation of Reagents

All components including the slides must be brought to room temperature (RT: 18 °C to 25 °C) before use for at least 30 min. All liquid components must be mixed gently to ensure homogeneity.

3.1.1 Slides

The slides are individually sealed with a desiccant in aluminum bags. Remove slides carefully from the bags directly before use.

3.1.2 Calibrators

The calibrators are ready-to-use and must not be diluted any further.

3.1.3 Controls

The positive and the negative controls are ready-to-use and must not be diluted any further. Controls must be used in each test run. Laboratories can also validate their own control samples and use them alternatively.

3.1.4 Sample Diluent

The sample diluent is ready-to-use.

3.1.5 PBS Buffer

A PBS buffer mixture must be dissolved in 1.0 L distilled water by stirring or shaking before use. The resulting washing solution should have a pH of 7.4 ± 0.2 and is stable for four weeks when stored at 2 °C to 8 °C. Do not use if the pH changes, the solution turns cloudy, or if precipitates form.

3.1.6 Conjugate

The conjugate is ready-to-use. Exposure to strong light should be avoided.

3.1.7 Mounting Medium

The mounting medium is ready-to-use.

3.2 Preparation of Samples

3.2.1 Sample Material

The use of freshly collected serum from blood taken by venipuncture is recommended. The use of icteric, lipemic, hemolytic or bacterially contaminated samples should be avoided. Insoluble substances must be

removed from the sample by centrifugation. Samples must not be thermally inactivated.

3.2.2 Sample Dilution

The samples must be diluted 1:20 (e. g. 10 µL + 190 µL) with sample diluent and mixed thoroughly. Building of foam should be avoided.

3.2.3 Sample Storage

Samples may be kept at 2 °C to 8 °C up to three days. Long-term storage requires -20 °C. Repeated freezing and thawing should be avoided. For multiple use, samples should be aliquoted and kept at -20 °C.

4 Test Performance

4.1 Procedure

A sufficient amount of slides must be prepared. Do not touch the wells and do not allow the substrate slides to dry during the entire test procedure. The indicated incubation times and temperatures must be adhered to and significant time shifts during pipetting samples and reagents must be avoided.

Step	Description
1. Addition of, controls and diluted samples	Add 50 µL of ready-to-use calibrators (in the direction CAL0, CAL1, CAL2, CAL3) onto the wells 1, 2, 3 und 4 of the first slide. Add 50 µL of ready-to-use controls and 50 µL of diluted sample per well. Make sure each well is completely covered
2. Incubation	Incubate for 30 min. at RT in a moist chamber
3. Wash cycle	Aspirate the liquid from each well using a vacuum pump and directly transfer the slide in the PBS bath. Discard the first PBS bath after transferring the last slide into the bath, fill in fresh buffer. After 2 min. change the PBS bath. All over the slides are washed 5 times in the PBS bath. Remove each slide individually, shake off PBS, and remove any still remaining liquid by carefully blotting with the paper templates.
4. Addition of conjugate	Add 50 µL ready-to-use conjugate to each well. Make sure each well is completely covered.
5. Incubation	Incubate for 30 min. at RT in moist chamber protected from direct light.
6. Wash cycle	Aspirate the liquid from each well using a vacuum pump and directly transfer the slide in the PBS bath. Discard the first PBS bath after transferring the last slide into the bath, fill in fresh buffer. After 2 min. change the PBS bath. All over the slides are washed 5 times in the PBS bath. Remove each slide individually, shake off PBS, and remove any still remaining liquid by carefully blotting with the paper templates.
7. Addition of mounting medium	Apply 1 drop per well ready-to-use mounting medium.
8. Coverage	Carefully place one coverslip onto the slide allowing the mounting medium to form a uniform layer between coverslip and slide. Avoid air bubbles. Wipe off excess mounting medium from the edges and the bottom of the slide using absorbent paper.
9. Analysis	Analyze the slide using the AKLIDES® or akiron® system with 490 nm excitation and 520 nm emission wavelengths within 3 hours.

4.2 Automation

Automated processing of the immunoassay must be performed analogous to manual use, e. g. by use of the AKENOMI® system or equivalents, and validated by the user.

5 Test Evaluation

5.1 Metrological Traceability

The immunoassay is calibrated using internal samples as references.

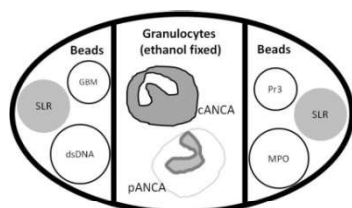
5.2 Evaluation

The AKLIDES® or akiron® system provides images for the qualitative evaluation of the fluorescence pattern to the operator. In addition, a suggestion for the calculated end titer is presented. The intensity of the fluorescence is expressed in artificial units (AU).

Furthermore, quantitative results for antibody activities directed against the antigens coated to the microspheres in U/mL or IU/mL are calculated.

Evaluated samples can be verified at the PC by the use of stored images.

The middle section of the wells contains ethanol fixed granulocytes, the right hand section of the wells contains different sized antigen coated beads: PR3 (Ø ca. 9 µm), MPO (Ø ca. 15 µm) and the „Size and Location Reference“ (SLR) which, with a diameter of 13 µm lie between the two antigen beads, aid in the differentiation between the two. The left hand section contains coated beads: GBM (Ø ca. 10 µm), dsDNA (Ø ca. 15 µm) and SLR. Due to a co-polymerized dye, the reference beads always show a green fluorescence.



Evaluation of the tissue or cells is performed according to fluorescence pattern and intensity. The fluorescence intensity may be semi-quantitated following the guidelines established by the CDC, Atlanta, USA:

4+ = maximal fluorescence, brilliant yellow-green

3+ = less brilliant yellow-green fluorescence

2+ = definite but dull yellow-green fluorescence

1+ = very dim subdued fluorescence

Evaluation of the beads is performed according to ring fluorescence intensity.

+ = visible ring fluorescence

- = no visible ring fluorescence

Combinations of fluorescence patterns

1. Cell pattern is cANCA and ring fluorescence of the PR3 beads.
Interpretation: PR3 antibody
2. Cell pattern pANCA and ring fluorescence of the MPO beads.
Interpretation: MPO antibody
3. Cell pattern negative and ring fluorescence of the GBM beads.
Interpretation: GBM antibody
4. Cell pattern nuclear/pANCA and ring fluorescence of the dsDNA beads.
Interpretation: dsDNA antibody
5. Cell pattern is negative, beads show a clear ring fluorescence
Interpretation: Test should be repeated with lower dilution. With titrations especially, positive fluorescence can remain for longer on the beads than on the cells.
TIP: Obtaining this result with the 1:20 screen, repeat with a 1:10 dilution to detect weak positive ANCA on the cells.
6. Cell pattern is positive, beads show no ring fluorescence
Interpretation: no ANCAs with PR3 or MPO specificities are present, possible presence of antibodies against other granulocyte antigens (e.g. lactoferrin, elastase, bactericidal/permeability increasing enzyme BPI) or other nuclear components (ANA) must be investigated further.
7. Cell pattern pANCA, no ring fluorescence on MPO beads
Interpretation: Presence of antibodies against nuclear components (ANA) is possible, for clarification perform further tests on formalin fixed granulocytes (AKLIDES® pANCA, REF 4072) and/or HEP-2 cells (AKLIDES® ANA plus, REF 4065).
8. Strong background signal in the bead regions
Interpretation: Ring fluorescence of the beads cannot be measured, and so this sample must be tested again.

The calibration of the test for the quantitative analysis is carried out with the help of four calibrators (CAL0, CAL1, CAL2, CAL3), which has to be in each staining preparation. The four calibrators are used by the software to adapt the calibration curve for the compensation of staining variations. The calibration curve is provided in each batch certificate, in which the curve is ciphered in a barcode for the AKLIDES® system. When

using a batch for the first time the AKLIDES® system asked for scanning the barcode so that the respective batch specification is loaded.

5.3 Criteria of Validity

Test runs are only valid if the following criteria of validity are fulfilled:

- The negative control must be evaluated negative.
- The positive control must be evaluated positive as indicated on the quality control certificate.

If these criteria are not met, the test is not valid and must be repeated.

5.4 Troubleshooting

In case of an invalid test run, the expiry dates and storage conditions, incubation times and temperatures, and precise calibration of all instruments used should be verified. If no reason for an invalid test run could be identified, please contact the supplier or manufacturer of the product. A troubleshooting guide is available upon request.

5.5 Reference Ranges

The reference ranges are indicated below:

	Interpretation
Granulocyte antibody titer < 1:20	negative
Granulocyte antibody titer ≥ 1:20	positive
PR3 antibody activity ≤ 4.5 IU/mL	negative
PR3 antibody activity 4.5 – 5.0 IU/mL	borderline
PR3 antibody activity > 5.0 IU/mL	positive
MPO antibody activity ≤ 4.5 IU/mL	negative
MPO antibody activity 4.5 – 5.0 IU/mL	borderline
MPO antibody activity > 5.0 IU/mL	positive
GBM antibody activity ≤ 6.0 U/mL	negative
GBM antibody activity 6.0 – 7.0 U/mL	borderline
GBM antibody activity > 7.0 U/mL	positive
dsDNA antibody activity < 9.5 IU/mL	negative
dsDNA antibody activity 9.5 – 10.0 IU/mL	borderline
dsDNA antibody activity > 10.0 IU/mL	positive

As a result of different seroprevalences in individual regions, each laboratory should verify the reference ranges by own analysis and adapt, if necessary.

5.6 Interpretation of Test Results

A positive test result indicates the presence of specific antibodies. A negative result indicates the absence of specific antibodies, but does not exclude the possibility of an autoimmune reaction. In case of a borderline test result, a reliable evaluation is not possible.

5.7 Limitations of the Method

The intensity of the fluorescence does not always reflect the antibody activity in the sample.

Samples and wash solutions contaminated with bacteria or fungi could cause unspecific staining.

Proteolytic enzymes in patient samples could result in damage or loss of the cell substrate fixed on the slide or of the antigens coated on beads.

The interpretation of test results must always be considered in combination with the clinical picture of the patient. The diagnosis should not be based on the results of a sole diagnostic method. All clinical and laboratory findings should be evaluated to state a diagnosis. For confirmation, further investigations should be carried out.

6 Performance Characteristics

6.1 Analytical Performance Characteristics

6.1.1 Analytical Sensitivity and Specificity

No cross-reactivities of antibodies against similar antigens have been observed.

6.1.2 Precision

The precision of test results was assessed by the determination of the intra- and interassay variation by the analysis of multiple samples with different antibody activities. No differences in the qualitative evaluation have been detected.

6.2 Diagnostic Performance Characteristics

6.2.1 Diagnostic Sensitivity and Specificity

Sensitivity and specificity of the Bead populations were detected by measurement of 293 samples (40 GPA, 38 MPA, 10 AAV, 2 EGPA, 48 SLE, 43 Good pasture/a-GBM, 55 healthy controls):

	Sensitivity	Specificity
PR3	85.4 %	97.6 %
MPO	73.5 %	95.5 %
GBM	88.4 %	96.8 %
dsDNA	83.3 %	94.7 %

7 Warnings and Precautions

The product is designed exclusively for *in vitro* diagnostic use by qualified, authorized and trained personnel. All test components and human samples should be handled with care as potentially hazardous. Good laboratory practices (GLP) and all relevant regulations should be adhered to.

In case the product is damaged or product information including labelling is wrong or incorrect, please contact the manufacturer or supplier.

This product contains preparations of human and / or animal origin. Any material derived from human body fluids or organs used for the preparation of components were tested and found negative for HBsAg (Hepatitis B-Virus-surface Antigen) and anti-HIV as well as anti-HCV antibodies. However, all components and all patient samples should be handled as potentially hazardous in accordance with national laws and appropriate guidelines on biological safety.

As the product contains potentially hazardous materials, the following precautions should be followed: Do not smoke, eat or drink while handling kit material or samples. Avoid direct contact to kit material or samples by wearing protective gloves laboratory coat and safety glasses. Never pipette material by mouth. Wipe up spills promptly and wash the affected surface thoroughly with a decontaminant. Wash hands thoroughly after use.

Some of the reagents contain sodium azide (< 0.1 %) as a preservative and must not be swallowed or allowed to come into contact with skin or mucosa. The possible formation of heavy metal azides in the drainage has to be prevented by sufficient rinsing with water.

Some of the reagents contain DAPI (5 µg/mL) as a staining agent and must not be swallowed or allowed to come into contact with skin or mucosa.

The information in the safety data sheet on possible hazards, first aid measures, measures in the event of the unintentional release of large quantities, handling and storage, personal protective equipment, information on disposal as well as information on toxicology must be observed.

Any serious incident that has occurred in relation to the device shall be reported to the manufacturer and the competent authority of the member state in which the user and/or the patient is established.

8 Disposal

For decontamination and disposal the recommendations of the CDC as well as the relevant local and national environmental guidelines and regulations should be adhered to. Samples, potentially contaminated materials and infectious waste must be decontaminated, e.g. by autoclaving for 20 min. at 121 °C.

9 References

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10 Symbols

	Manufacturer
	CE marking of conformity
	<i>In vitro</i> diagnostic medical device
	Catalogue number
	Unique device identifier
	Batch code
	Temperature limit
	Use-by date
	Consult instructions for use
	Contains sufficient for <n> tests
	Do not re-use
	Caution
	Biological risk
	Keep away from sunlight
	Slide with x wells
	Calibrator
	Negative control
	Positive control
	Sample diluent
	Conjugate
	Wash buffer
	Mounting medium
	Blotting templates
	Coverslip

11 Changes

Changes in current Instructions for Use

Current Version	006/08.2022
Summary of Changes	Editorial changes and adaption of PBS buffer in use stability.